TELEMETRI GECIDI

Ana karttan Modbus RTU okur, MQTT ile aga tasir

AYRI KART: radyo, olcum kartinin uzerinde degil

01-guc-rs485

Guc ve izole RS-485 - panodan Modbus

Dosya: 01-guc-rs485.kicad_sch

02-esp32

ESP32-S3 - MQTT gecidi

Dosya: 02-esp32.kicad_sch

Melike Atılgan

Sheet: /
File: telemetri-gecidi.kicad_sch

Title: Telemetri Gecidi

Size: A3	Date: 2026-10-11	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 1/3

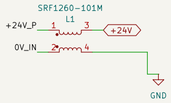
PANO BAGLANTISI



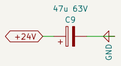


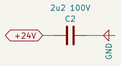


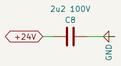






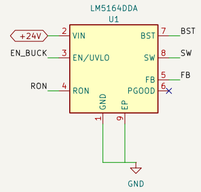








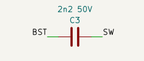
24 V –> 3,3 V SENKRON BUCK

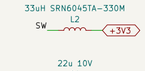


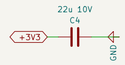


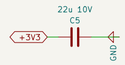


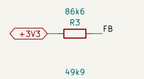












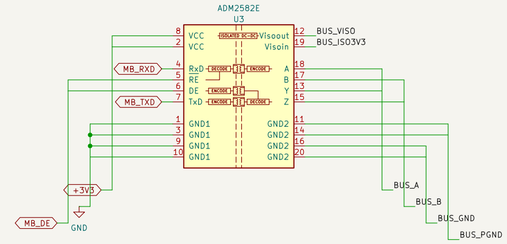


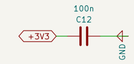




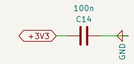


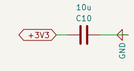
IZOLE RS-485 ADM2582E







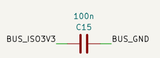








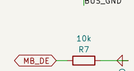


















TASARIM NOTLARI

1) BU KART AYRI. Radyo, olcum kartinin uzerinde DEGIL. ESP32 verici tepesinde ~350 mA darbeli akim ceker; ayni kartta 45 dB dinamik aralik isteyen CT on ucu olsaydi bu darbeler ADC referansina binerdi.

2) MODBUS SALT OKUNUR. Gecidin firmware i yalnizca okuma istegi gonderecek: kesme yetkisi olan bir cihazi aga acmak saldiri yuzeyi olur.

3) 3,3 V DOGRUDAN BUCK TAN. ESP32 verici tepesi ~0,5 A, izole RS-485 ~0,2 A; toplam ~0,7 A. 500 mA lik LDO buna yetmez; U1 in siniri 1 A.
BUCK (LM5164): Vout = 1.2 V x (1 + 86k6/49k9) = 3,28 V (+0,03 V dalgalanma agi):
Rron 27k4 –> 301 kHz; 33 uH –> 0,29 A dalgalanma; UVLO 16,9 / 15,7 V.
Giris: SMBJ33CA en cok 53,3 V ta kirpar, U1 in siniri 100 V, D2 100 V.
L1 SRF1260-101M: sargilar 1-3 ve 2-4 (uretici foyu).

5) VISOOUT –> VISOIN ve donusturucu topragi (11, 14) –> hat topragi (16, 20) ferritle (ADI AN-1349): ferritlerin altinda bakir yok.
SM712 1 A darbede +19 / −11 V; ADM2582E hat siniri −9 / +14 V. Darbe testi yok.

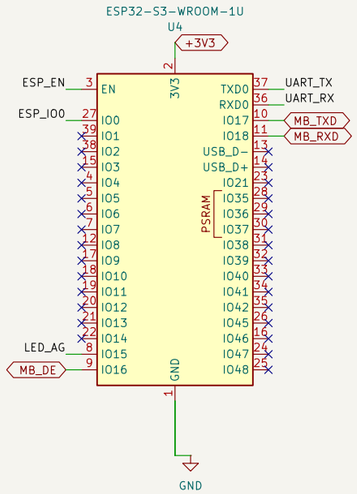
4) 120R sonlandirma: gecit hat sonudur, kalici takili.

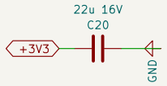
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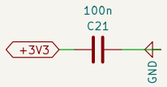
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Size: A2	Date: 2026-10-11	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 2/3

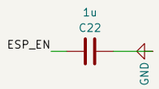
ESP32-S3-WROOM-1U





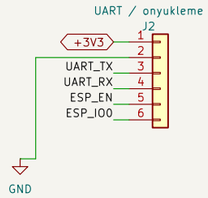




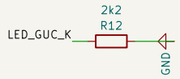




PROGRAMLAMA VE GOSTERGE









TASARIM NOTLARI

1) MODUL 'U' VARYANTI: ESP32-S3-WROOM-1U, PCB anten yerine IPEX konnektorlu. Anten pigtail ile PANONUN DISINA cikarilir.

   Metal pano radyo isaretini zayiflatir; o yuzden harici anten.

   Sematikte WROOM-1 sembolu kullanildi (pin uyumlu), ayak izi 1U.
2) ANTEN YOLU KARTTA YOK. IPEX modulun uzerinde; kartta 50 ohm iz,

   anten keep-out veya SMA gerekmez; kartta RF izi olmadigi icin

   empedans kontrolu de gerekmiyor.
3) EN pininde 10k + 1u: acilis sirasinda besleme oturana kadar

   moduldeki reset tutulur (ESP32 nin bilinen acilis gereksinimi).

4) IO0 10k ile YUKARI cekili: normal acilis. Programlama icin J2 uzerinden asagi cekilir.

5) UART0 (TXD0/RXD0) yalnizca programlama ve gunluk icin. Modbus

   trafigi AYRI UART ta (IO17/IO18); iki is ayni hatti paylasmiyor.

Melike Atılgan		
Sheet: /02-esp32/ File: 02-esp32.kicad_sch		
Title: ESP32-S3 – MQTT gecidi		
Size: A3	Date: 2026-10-11	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 3/3